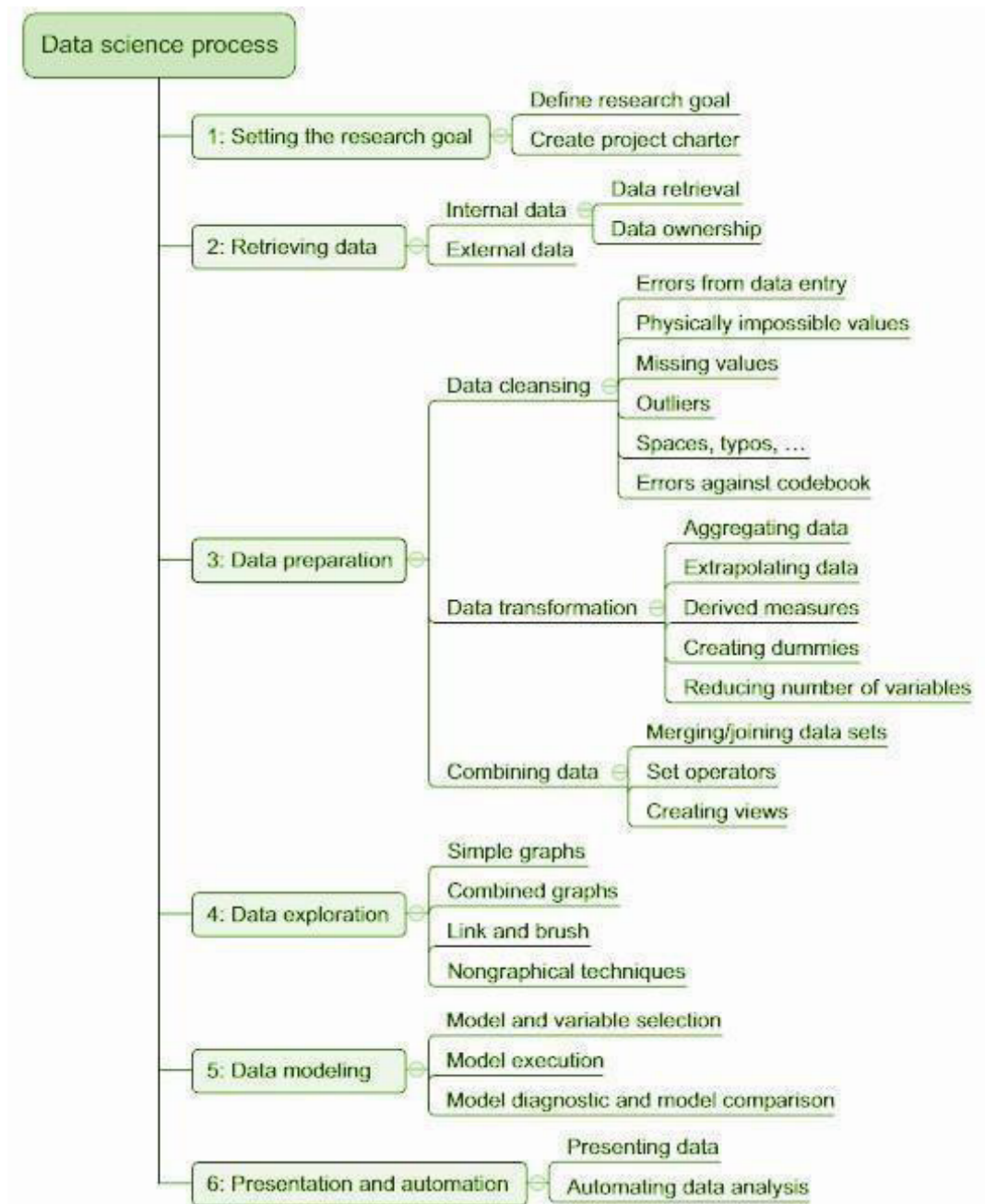




Step 1. Setting the Research Goal

Problem Definition

The research goal begins with **problem formulation**. The problem must be stated in a way that can be addressed using data.

- Convert real-world problems into **data-driven questions**
- Avoid vague or broad problem statements
- Focus on *what decision or prediction is required*

Example:

- Business question: *Why are sales decreasing?*
- Data science goal: *Identify key factors affecting monthly sales using historical data*

Analytical Objective

Once the problem is defined, the **analytical objective** must be identified. This determines the type of analysis to be performed.

Common objectives include:

- **Descriptive** – What happened?
- **Diagnostic** – Why did it happen?
- **Predictive** – What will happen?
- **Prescriptive** – What should be done?

Understanding the Domain Context

Domain understanding is essential for defining meaningful goals.

- Provides background knowledge of the problem
- Helps identify relevant variables
- Prevents misinterpretation of results

For example:

- In healthcare, clinical knowledge is required to define valid health indicators
- In education, understanding grading systems is necessary to define performance metrics

Identifying Stakeholders and Decision Makers

A research goal must consider **who will use the results**.

Stakeholders may include:

- Managers
- Policymakers
- Teachers or doctors
- End users

The goal should be framed to support **decision-making**, not just analysis.

Defining Success Criteria

Success criteria determine **how the outcome will be evaluated**.

Examples:

- Accuracy, precision, recall
- Reduction in cost or error
- Improved efficiency or performance

Assumptions, Constraints, and Ethics

The research goal must acknowledge:

- Data availability

- Time and resource constraints
- Privacy and ethical considerations
- Legal and organizational limitations

Create project charter: Outline objectives, scope, timeline, data sources, and expected outcomes.

The final outcome of Step 1 is a **project charter**, which formally documents:

- Problem statement
- Objectives
- Scope
- Data sources
- Expected deliverables

This charter acts as a **roadmap** for the entire data science project.

Step 2. Retrieving Data

After clearly defining the research goal, the next step in the data science process is Retrieving Data. This step focuses on collecting relevant, reliable, and usable data required to achieve the defined objective.

- **Internal data:** Data from within the organization (databases, records).
- **External data:** Data from outside sources (websites, surveys, APIs).
- **Data retrieval:** Extracting data from files, databases, or online sources.

- **Data ownership:** Understanding who owns the data and permission to use it.

1. Identifying Data Requirements

Before collecting data, the data scientist must determine:

- What type of data is needed (numerical, categorical, text, time-series)
- Which variables are relevant to the research goal
- The level of detail (granularity) required

Example:

For predicting student performance:

- Required data: attendance, internal marks, assignment scores
- Not required: student address or phone number

2. Sources of Data

A. Internal Data

Data generated within an organization.

Examples:

- Student academic records
- Hospital patient databases
- Company sales transactions

Advantages:

- High reliability

- Easy access
- Better understanding of data structure

B. External Data

Data obtained from outside sources.

Examples:

- Government open data portals
- Surveys and questionnaires
- Web data, APIs, social media

Challenges:

- Data quality may vary
- Legal and ethical issues
- Data format inconsistencies

3. Data Retrieval Methods

Data can be collected using different techniques:

- **Databases:** SQL queries
- **Files:** CSV, Excel, JSON, XML
- **APIs:** REST APIs for live data
- **Web scraping:** Extracting data from websites
- **Sensors and logs:** IoT devices, system logs

The method chosen depends on data availability and project constraints.

4. Data Ownership and Access Rights

Before retrieving data, it is essential to:

- Identify the data owner
- Obtain necessary permissions
- Ensure compliance with privacy laws and institutional policies

5. Data Quality Considerations at Collection Stage

At the time of retrieval, the following should be assessed:

- Completeness
- Accuracy
- Consistency
- Timeliness

Collecting poor-quality data increases effort in later cleaning stages.

6. Metadata and Documentation

Metadata describes the data, such as:

- Variable meaning
- Units of measurement
- Data collection method

Proper documentation helps avoid misinterpretation during analysis.

7. Challenges in Data Retrieval

Common issues include:

- Missing or incomplete data
- Data stored in multiple formats
- Restricted or sensitive data
- Large data volume

Step 3. Data Preparation

This is the most time-consuming but crucial step.

A. Data Cleansing

Fix errors and improve data quality.

1. Errors from Data Entry

These errors occur when data is entered manually and mistakes are made.

Example:

- Marks entered as **880 instead of 88**
- Name typed as **“Jonh” instead of “John”**

Impact:

Leads to incorrect analysis and misleading results.

2. Physically Impossible Values

These are values that **cannot exist in real life**.

Example:

- Age = **-5 years**

- Height = **350 cm** for a human
- Marks = **120 out of 100**

Impact:

Violates real-world logic and must be corrected or removed.

3. Missing Values

Missing values occur when **no data is recorded** for a variable.

Example:

- Student attendance is blank
- Patient's blood pressure not recorded

Handling:

- Replace with mean/median
- Use previous value
- Remove the record

4. Outliers (Extreme Values)

Outliers are values that are **much higher or lower** than most data points.

Example:

- Salaries: 25,000; 27,000; 28,000; **5,00,000**
- Exam scores: 45, 50, 55, **0** (due to absence)

Impact:

Can distort averages and model performance.

5. Spaces, Spelling Mistakes, and Typos

These are **formatting and text errors** in categorical data.

Example:

- “CSE”, “ cse ”, “C.S.E” treated as different values
- City names: “Chennai” vs “Chennnai”

Impact:

Causes incorrect grouping and counting.

6. Errors Against the Codebook (Rules of Data)

A codebook defines **valid values and formats** for each variable.

Example:

- Gender codebook: {M, F} value entered as “Male”
- Department codebook: {CSE, ECE, MECH} value entered as “IT”

B. Data Transformation

Data transformation involves **modifying and restructuring data** so that it becomes suitable for analysis and modeling.

1. Aggregating Data (Daily Monthly)

Aggregation means **summarizing detailed data into a higher level**.

Example:

- Daily sales Monthly total sales
- Hourly temperature Daily average temperature

Why needed:

- Reduces data size
- Reveals overall trends
- Makes analysis easier and faster

2. Extrapolating Data

Extrapolation is the process of **estimating unknown or future values** based on existing data patterns.

Example:

- Predict next month's sales using past sales data
- Estimate missing sensor readings

Why needed:

- Helps in forecasting
- Handles incomplete datasets

3. Creating Derived Measures (BMI, Averages)

Derived measures are **new variables created from existing variables**.

Example:

- $BMI = \text{Weight} / (\text{Height}^2)$
- $\text{Average marks} = (\text{Test1} + \text{Test2} + \text{Test3}) / 3$

Why needed:

- Provides meaningful indicators
- Improves model accuracy
- Reduces complexity by summarizing information

4. Creating Dummy Variables (Yes/No 1/0)

Dummy variables convert **categorical data into numerical form**.

Example:

- Gender: Male/Female 1/0
- Loan approval: Yes/No 1/0

Why needed:

- Most machine learning models require numerical input
- Enables mathematical operations

5. Reducing Number of Variables (Feature Selection)

Feature selection involves **removing irrelevant or redundant variables**.

Example:

- Removing ID or phone number from prediction models
- Keeping only important features like age and income

Why needed:

- Improves model performance
- Reduces overfitting
- Saves computation time

C. Combining Data

Combining data means **integrating data from multiple sources** to obtain a complete and consistent dataset.

1. Merging / Joining Datasets

Merging or joining combines datasets based on a **common key**.

Example:

- Student details table + marks table joined using Student ID

Why needed:

- Brings related information together
- Enables richer analysis

2. Set Operators (Union, Intersection)

- **Union:** Combines rows from two datasets without duplication
- **Intersection:** Keeps only common records between datasets

Example:

- Union: Students from Class A Class B
- Intersection: Students common to both sports and cultural teams

Why needed:

- Useful when working with similar structured datasets
- Helps compare and consolidate data

3. Creating Views (Logical Tables)

A view is a **virtual table** created using a query without storing data physically.

Example:

- View showing only final-year students and their GPA

Why needed:

- Simplifies complex queries
- Improves security by restricting data access

4. Data Exploration

Data Exploration is the process of understanding the structure, patterns, and characteristics of data before applying any statistical or machine learning models. It helps in identifying trends, relationships, errors, and anomalies in the dataset.

1. Simple Graphs

Simple graphs represent **one variable at a time** and help in understanding basic data distribution.

- **Bar charts** are used to compare values across different categories, such as the number of students in each department.
- **Line graphs** show trends or changes over time, such as monthly sales or yearly rainfall.
- **Histograms** display the frequency distribution of numerical data and help identify skewness and spread.

Purpose:

To visualize data clearly and identify patterns or unusual values.

2. Combined Graphs

Combined graphs display **two or more variables in a single graph** to study relationships.

Examples:

- Sales vs profit on the same chart
- Marks vs attendance using scatter plots

Purpose:

To understand how variables influence each other and detect correlations.

3. Link and Brush (Interactive Visualization)

Link and brush is an **interactive data visualization technique** where selecting data in one graph highlights the same data in other graphs.

Example:

- Selecting students with low marks in a scatter plot highlights the same students in a bar chart

Purpose:

To explore multidimensional data interactively and discover hidden relationships.

4. Non-Graphical Techniques

These techniques use **numerical summaries** instead of visuals.

- **Mean** represents the average value of data
- **Median** represents the middle value
- **Standard deviation** measures data variability or spread

Purpose:

To summarize data numerically and support graphical analysis.

5. Data Modeling

Data Modeling is the stage where mathematical and statistical models are built to answer the research question or solve the defined problem. In this step, patterns identified during data exploration are converted into predictive or descriptive models.**Model and variable selection:** Choose suitable algorithms and features

1. Model and Variable Selection

In this phase, suitable **models (algorithms)** and **input variables (features)** are chosen based on the nature of the problem and data.

- The type of problem (classification, regression, clustering) determines the model selection.
- Relevant variables are selected, and irrelevant or redundant variables are removed.
- Domain knowledge and exploratory analysis guide this selection.

Example:

- Linear regression for predicting marks
- Decision tree for pass/fail classification

Purpose:

To ensure the model is appropriate, simple, and effective.

2. Model Execution

Model execution involves **training the selected model using prepared data**.

- Data is divided into training and testing sets.
- The model learns patterns and relationships from the training data.
- Parameters are estimated during this process.

Example:

- Training a regression model using attendance and internal marks

Purpose:

To enable the model to learn from data and make predictions.

3. Model Diagnostics and Comparison

After training, the model's performance is evaluated.

- Accuracy, precision, recall, or error metrics are calculated.
- Residual analysis and validation techniques are applied.
- Multiple models may be compared to select the best-performing one.

Example:

- Comparing decision tree and logistic regression accuracy

Purpose:

To ensure reliability, avoid overfitting, and select the best model.

6. Presentation and Automation

Presentation and Automation is the final step in the data science process, where insights obtained from data modeling are **communicated clearly** to stakeholders and the analysis process is **made repeatable and efficient**.

1. Presenting Data

Presenting data involves **communicating results in an understandable and actionable form**.

- **Reports** summarize objectives, methods, findings, and recommendations in a structured manner.
- **Dashboards** provide interactive, real-time views of key metrics and performance indicators.
- **Charts and visualizations** such as bar charts, line graphs, and pie charts help in quick interpretation.
- **Conclusions and insights** explain what the results mean and how they support decision-making.

Purpose:

To ensure that technical results are easily understood by non-technical users and decision-makers.

2. Automating Analysis

Automating analysis involves creating **scripts, workflows, and pipelines** to repeat the analysis automatically.

- Scripts automate data cleaning, transformation, and modeling tasks.
- Pipelines ensure the same steps are executed consistently whenever new data arrives.
- Automation reduces manual effort, errors, and time consumption.

Example:

- Daily sales report generated automatically
- Automated model retraining with new data

Purpose:

To make the data science process efficient, scalable, and reproducible.